

ECON311, Winter 2024: Final Exam

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There are two parts in this exam. Part I has five questions and Part II has two questions for a total of 100 points. You must show all your work to get full marks. You are allowed a calculator. Good Luck!!!

1 Part I: Short-Answer Questions

Question I: Long-Run Growth

Consider the following Romer economy without capital:

$$\text{Production Function:} \quad Y_t = A_t L_{Y_t}$$

$$\text{Law of Motion for Productivity:} \quad A_{t+1} = A_t + \bar{z} A_t L_{A_t}$$

$$\text{Allocation of labour:} \quad L_{A_t} = \bar{l} \bar{L}, \quad L_{Y_t} = (1 - \bar{l}) \bar{L}$$

- (a) (6 points) Derive the rate of growth of output per person (g_y , where $y = \frac{Y}{L}$).
- (b) (6 points) Assume two countries, country 1 and country 2, that are the same in every respect, except across two dimensions. First, assume that in the year 1950, country 1 has a larger level of output productivity ($A_{1950}^{\text{country 1}} > A_{1950}^{\text{country 2}}$). Second, assume that country 2 is more productive in generating ideas ($\bar{z}^{\text{country 1}} < \bar{z}^{\text{country 2}}$). On the same graph draw the evolution of output per person over time for these two countries using a ratio scale and starting in the year 1950. Which country do you think will have larger output per person in the long run? Explain.

Question II: Cross-Country Income Differences (6 points)

As an economist working at the International Monetary Fund, you are given the following data for Burundi: observed per capita GDP, relative to the United States, is 0.01. Using the model of production ($Y_t = \bar{A} K^{1/3} L_t^{2/3}$) and assuming equal TFP ($\bar{A}$), predicted relative per capita GDP, given by $\frac{Y_t}{L_t} = \left(\frac{K_t}{L_t}\right)^{1/3}$, is 0.18. If we relax the assumption of equal TFP, what is the total factor productivity of Burundi relative to the US according to the model of production?

Question III: International Transmission of Monetary Policy (6 points)

Suppose that the Federal Reserve in the US decides to implement expansionary monetary policy due to a recession ($\downarrow i^{US}$). Assuming that the Canadian economy is at the potential output level, use the IS-MP model and the Phillips Curve to analyse how this policy affects the Canadian economy in the short run. To answer this question, you do not need to derive all the equations of the model.

However, you need to carefully discuss the effect of the policy on the value of the Canadian Dollar relative to the US Dollar and show the effect of this shock graphically (in both, the IS-MP and Phillips Curve graphs). Suppose that the Bank of Canada wants to return the economy to its potential output level, how would it use domestic monetary policy to counter this shock?

Question IV: Financial Crisis (6 points)

Suppose that the IS curve is as follows: $\tilde{Y}_t = \frac{1}{1-b_c} [\bar{a} - \bar{b}_i (R_t - \bar{r})] - 1$. Explain how you would modify this equation to account for financial frictions (i.e. risk in financial markets, $\bar{f}$). Use this equation and the IS-MP model to explain the effect of a financial crisis (i.e. an increase in risk in the financial sector, $\uparrow \bar{f}$).

Question V: Solow Growth Model (8 points)

The economy of the country of Atlantis is given by:

Production Function	$Y_t = \bar{A} (K_t)^{\frac{1}{3}} (\bar{L})^{\frac{2}{3}}$
Capital Accumulation	$\Delta K_{t+1} = I_t - \bar{d}K_t$
Investment	$I_t = \bar{s}Y_t$
Consumption	$C_t = (1 - \bar{s})Y_t$

Atlantis' GDP per-capita and consumption per-capita are very low and the government is contemplating different policies to increase their long-run levels. The government does not currently have an economist to guide their policies and, therefore, have decided to hire you to help them. The prime minister would like to know your assessment of the effect of an increase in the savings rate, $\bar{s}$. The prime minister thinks that, by increasing the savings rate, GDP per-capita and consumption per-capita will both decrease immediately, but increase in the long run. Is the prime minister right? To assess the policy, include in your discussion Solow diagrams with total output, and diagrams with time on the horizontal axis and GDP per-capita and consumption per-capita, separately, on the vertical axis.

2 Part II: Long-Answer Questions

Question I: IS-MP Model

Consider the following economy:

$$\text{Expenditure:} \quad E_t = C_t + I_t + G_t + X_t - M_t$$

$$\text{Consumption:} \quad C_t = a_c + b_c(Y_t - T_t)$$

$$\text{Investment:} \quad I_t = a_i - b_i R_t$$

$$\text{Tax:} \quad T_t = b_T Y_t$$

$$\text{Inflation:} \quad \pi_t = \pi_{t-1} + \bar{v} \tilde{Y}_t$$

$$\text{Exogenous Variables:} \quad G_t = \bar{G}, \quad M_t = \bar{M}, \quad X_t = b_X Y_t^F,$$

, where $0 < b_c < 1$ and $0 < b_T < 1$. In this economy, the level of exports depend on the income level of a foreign country, Y_t^F , which is taken as an exogenous parameter by the domestic country (i.e. b_X is **not** part of the expenditure multiplier, and $b_X Y_t^F$ should be treated as an exogenous variable similar to $\bar{G}$). Moreover, consumption depends on disposable income (income net of taxes) and not just on the level of income as we previously assumed. Use the previous equations to answer the following questions:

- (a) (7 points) Solve for the equilibrium in the goods market. What is the expenditure multiplier?
- (b) (7 points) Use the Keynesian Cross to carefully explain the effect of an exogenous increase in income of the foreign country, Y_t^F (carefully label the graph).
- (c) (5 points) Derive the IS curve using deviations from potential output as the endogenous variable. For this question assume that there is a long-run level of output in the foreign country, denoted $\bar{Y}^F$, that is consistent with potential output in the domestic economy.
- (d) (10 points) Suppose the economy is at its potential output equilibrium with a constant 2% inflation rate. Assume that, starting in the year 2000, there is a recession in the foreign country (i.e. Y_{2000}^F decreases from $\bar{Y}^F$) lasting until 2003. This recession reduces the inflation rate in the **domestic** country by 0.5 percentage points per year (a total of 2 percentage point decrease). Assume that in 2006, the government decides to use monetary policy to return inflation back to the 2% target by increasing inflation by the same amount for two consecutive years. Using the full short-run model (IS-MP and Phillips curve), explain what happens to the economy **over time**. Be sure to include graphs showing how output and inflation respond over time (carefully label all graphs).
- (e) (7 points) Net exports, denoted by $NX_t = X_t - M_t$, model the economy's foreign trade. A positive value of NX_t denotes a trade surplus and a negative value denotes a trade deficit. Suppose that there is balanced trade when the economy is in its long-run equilibrium ($NX_t = b_X \bar{Y}_t^F - \bar{M} = 0$). Show how net exports change across time with the shock in part (d) and subsequent government response.

Question II: The AS-AD Model

Consider the following economy:

Aggregate Expenditure	$E_t = C_t + I_t + G_t + X_t - M_t$
Consumption	$C_t = a_c + b_c Y_t$
Investment	$I_t = a_i - b_i R_t$
Inflation	$\pi_t = \pi_t^e + \bar{v} \tilde{Y}_t + \bar{o}_t$
Policy Rule	$R_t - \bar{r} = \bar{m}(\pi_t - \bar{\pi})$
Government Expenditure and Trade	$G_t = \bar{G}, \quad M_t = \bar{M}, \quad X_t = \bar{X}$

Use the previous equations to answer the following questions:

- (a) (8 points) Derive the AD curve using deviations from potential output.
- (b) (10 points) Assume that $\pi_t^e = \pi_{t-1}$. Suppose the economy is at its potential output equilibrium with a constant $\bar{\pi}$ inflation rate. Using the AS-AD model, carefully explain what happens to the economy **over time** if there is a **multiple period** negative demand shock ($\bar{a}' < (1 - b_c)$). How would your answer to this question change if there was an increase in $\bar{m}$? Explain.
- (c) (8 points) Repeat the exercise in part b, but assume now that $\pi_t^e = \bar{\pi}$.